AfyaPlus triage: what it costs, and what we recommend

One-page brief for the CTO and Finance · 11 October 2026 · five-minute read

Recommendation: GO, with the five conditions at the bottom. One triage answer costs **$0.0031**, or **$3.09 per 1,000**, all-in. That is about **$185 a month** at today's 2,000 requests a day. The AI model itself is only 1% of that. Most of the cost is staff time ($150) and the always-on server ($34).

1. If traffic jumps tenfold

An outbreak month at 20,000 requests a day costs about **$241**, **$56 more** than a normal month. Each answer gets *cheaper* ($0.40 per 1,000), because the fixed costs are shared by more requests. Capacity is not the problem: one server copy handled about 31 requests a second in our load test, and a spike day peaks below 1. We run 1 copy normally, 2 in a spike, and **never more than 3**. That is the spending ceiling.

2. Two savings we built and measured

What it does	Measured result	The honest catch
Remembers answers to messages it has already seen in the last 10 minutes	On our 100-message test, 60% were answered from memory. AI cost fell 60%. Typical wait went from 1.0 s to instant.	The slowest answers stay about as slow (a new question still asks the AI). Real repeat rates must be measured before we quote this saving.
Handles after-hours messages in groups of 4, ready for the 08:00 review	AI cost for those messages fell 32%. 96% of answers matched answering one at a time.	Bigger groups saved more but downgraded some urgencies, so we chose 4. Danger words (chest pain, bleeding) are always flagged urgent.

Because the AI is a small part of the bill, these savings matter mostly for speed, and as insurance if AI prices rise. At a 17× higher AI price (gpt-4o), remembering answers would save about $223 in a spike month.

3. Should we run the AI on our own servers?

Not now. Renting our own AI server costs **$834 to $3,861 a month** whether or not anyone uses it, against about $2 a month for the AI service today. It only pays off at **433× today's traffic**. We would reconsider if traffic grew enormously, if prices jumped, or if a data-protection ruling required patient messages to stay in-country. Even then, a hosted service in-region comes before owning servers.

4. The main risk, and what we do about it

Risk: a partner app that keeps resending requests (a “retry loop”). It adds about a fifth to the AI bill and can lock out real clinicians. **Already in place:** a per-user request limit, at most one automatic retry, and the 3-copy ceiling. **Alert:** cloud spend is capped at $80 a month, and a warning at $64 (80%) goes to Finance and the on-call engineer. That warning stays silent in a normal month. The pre-agreed response is to tighten that partner's request limit, a change that takes minutes.

5. Go, on these conditions

1. Apply the $80 cloud budget and alerts when the cloud account is opened (written and checked, not yet live).
2. The owner of the AI account sets a $25 monthly limit with OpenAI. The cloud budget cannot see the AI bill.
3. Service target: 95 of every 100 answers within 2.5 seconds (“p95”). Measured today: 1.5 s without memory. Reviewed monthly.
4. Re-measure the real repeat rate and after-hours agreement in month one, before any saving is reported upward.
5. If the AI service is down, the app says so and points to the clinic. It never invents advice.

Figures come from 100+ live test requests on 11 Oct 2026 and published Azure and OpenAI prices; staff time ($25/h × 6 h a month) is an assumption. A shared memory store (Redis) is built and tested but held back until about 2.6 M requests a month, when it starts paying for itself. Details: cost/cost_model.md, levers/measurements.md, memo/api_vs_selfhost.md.